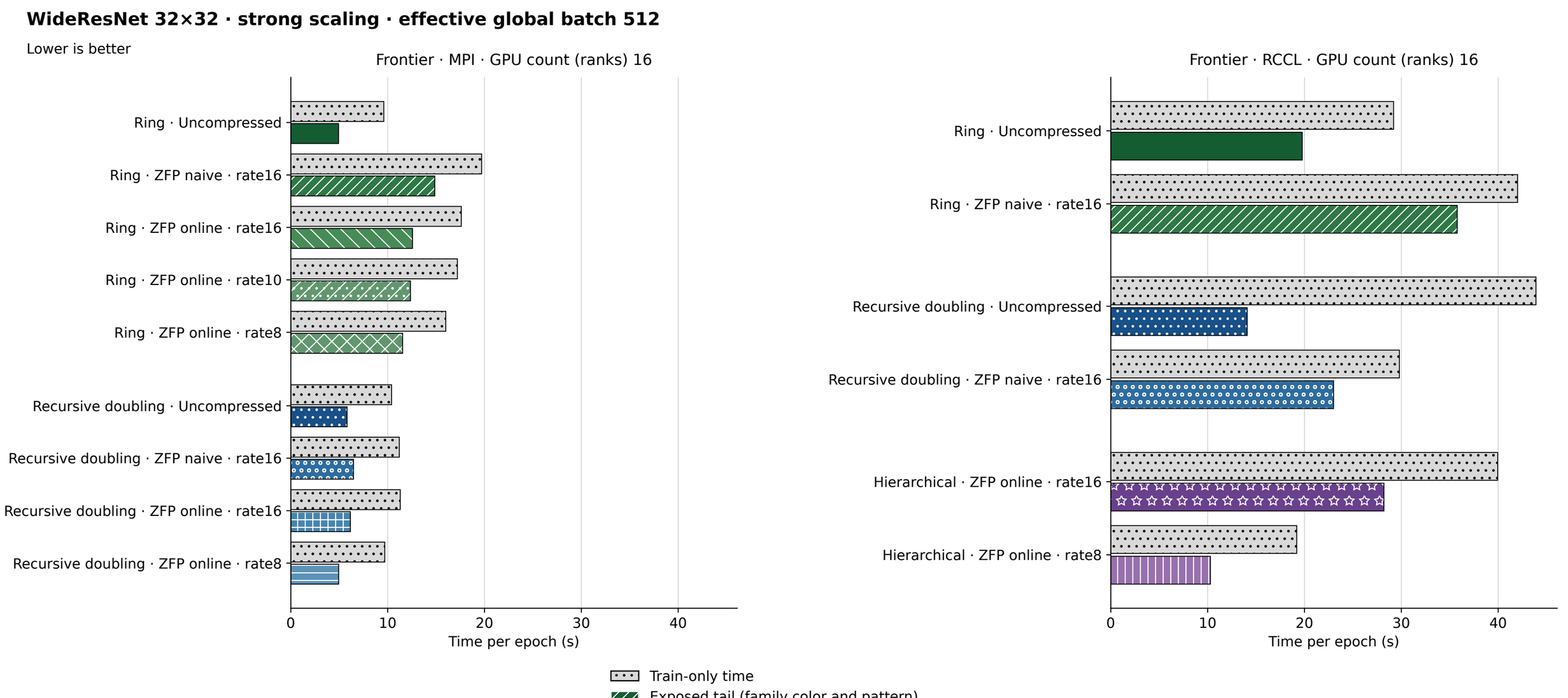

WideResNet 32×32 · strong scaling · effective global batch 512
Lower is better
Frontier · MPI · GPU count (ranks) 16
Ring · Uncompressed
Ring · ZFP naive · rate16
Ring · ZFP online · rate16
Ring · ZFP online · rate10
Ring · ZFP online · rate8
Recursive doubling · Uncompressed
Recursive doubling · ZFP naive · rate16
Recursive doubling · ZFP online · rate16
Recursive doubling · ZFP online · rate8
0
10
20
30
40
Time per epoch (s)
Frontier · RCCL · GPU count (ranks) 16
Ring · Uncompressed
Ring · ZFP naive · rate16
Recursive doubling · Uncompressed
Recursive doubling · ZFP naive · rate16
Hierarchical · ZFP online · rate16
Hierarchical · ZFP online · rate8
0
10
20
30
40
Time per epoch (s)
Train-only time
Exposed tail (family color and pattern)
Epoch 1 excluded · missing metrics are not zero · r = ZFP rate